NICHANAMETLA PRANEETH

AI & Robotics Undergraduate

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ABOUT ME

AI & Robotics undergraduate focused on learning machine learning, robotics systems, and intelligent automation. Currently expanding expertise in Python and AI technologies. Interested in developing innovative AI-driven solutions and contributing to advanced robotics research.

EDUCATION

B.Tech in Artificial Intelligence & Robotics
Dayananda Sagar University, Bengaluru
2024 – 2028

- Studying core concepts in Artificial Intelligence, Machine Learning, and Robotics systems
- Developing programming skills in C and Python for problem solving
- Exploring AI-driven projects and robotics simulations

PROJECTS

PLC-Based Industrial Automation System

- Designed an industrial automation system using PLC and ladder logic programming.
- Implemented control logic to automate and monitor basic industrial operations.

Shoe_Classifier (DL)

- Developed a deep learning-based shoe classification system capable of identifying different types of sports footwear using image data.
- Implemented model training, evaluation, and visualization techniques (including Grad-CAM) to improve accuracy and interpret model predictions.

Machine Learning Prediction Model

- Developed a prediction model using Python, Pandas, NumPy, and Scikit-learn.
- Performed data preprocessing, model training, and evaluation to analyze prediction accuracy.

TECHNICAL SKILLS

- Programming Languages:
  C, Python
- Libraries:
  NumPy, Pandas, Matplotlib, Scikit-learn
- Tools & Technologies:
  MATLAB, VS Code, Git, Google Colab

CERTIFICATIONS

- Introduction to Cybersecurity – Cisco Networking Academy
- PLC Programming & Industrial Automation – Rexroth Bosch
- Hydraulics and Pneumatics Industrial Automation - Rexroth Bosch

GITHUB PROJECTS

Shoe Classifier (DL)

- Developed a deep learning–based shoe classification system capable of identifying different types of sports footwear using image data.

Cleaning Robot Coverage System | Python, NumPy, Matplotlib

- Developed a Python-based Cleaning Robot Coverage 2D Simulator implementing Sweep, Spiral + BFS, and Random Bounce algorithms with performance analysis and battery-aware autonomous docking.